

INVENTORY AND SUPPLIER MANAGEMENT SYSTEM

Presented By : Wai Yan Lynn 67111411
Min Hein Thu 67111391
Pyae Sone 67111336



SYSTEM OVERVIEW



Manage stock levels and item movement transactions.

Create, approve, issue and receive supplier orders.

Generate low stock alerts when item quantity reaches the threshold.

Produce operational and management reports.

Control user accounts, roles and access rights.

SCOPE DEFINITION

IN SCOPE

- Stock Tracking
- Supplier Orders
- Low Stock Alerts
- Reporting
- User Management
- Audit Logs

OUT OF SCOPE

- Finance and accounting
- Logistics and delivery routing
- Supplier Portal
- POS Integration

SYSTEM MODULES



STOCK TRACKING



SUPPLIER ORDER



LOW STOCK ALERT

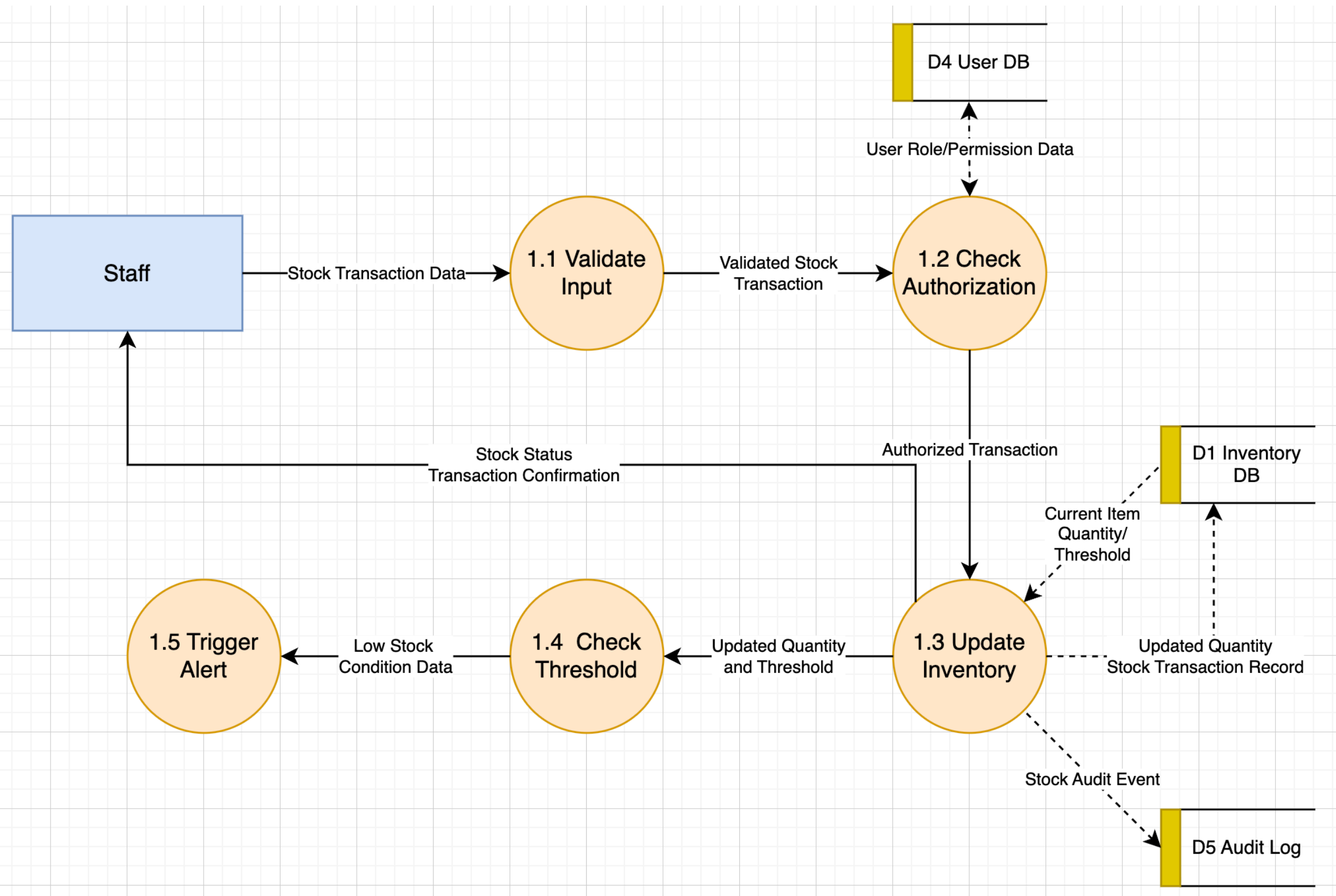


USER MANAGEMENT



REPORTING

1.0 STOCK TRACKING MODULE



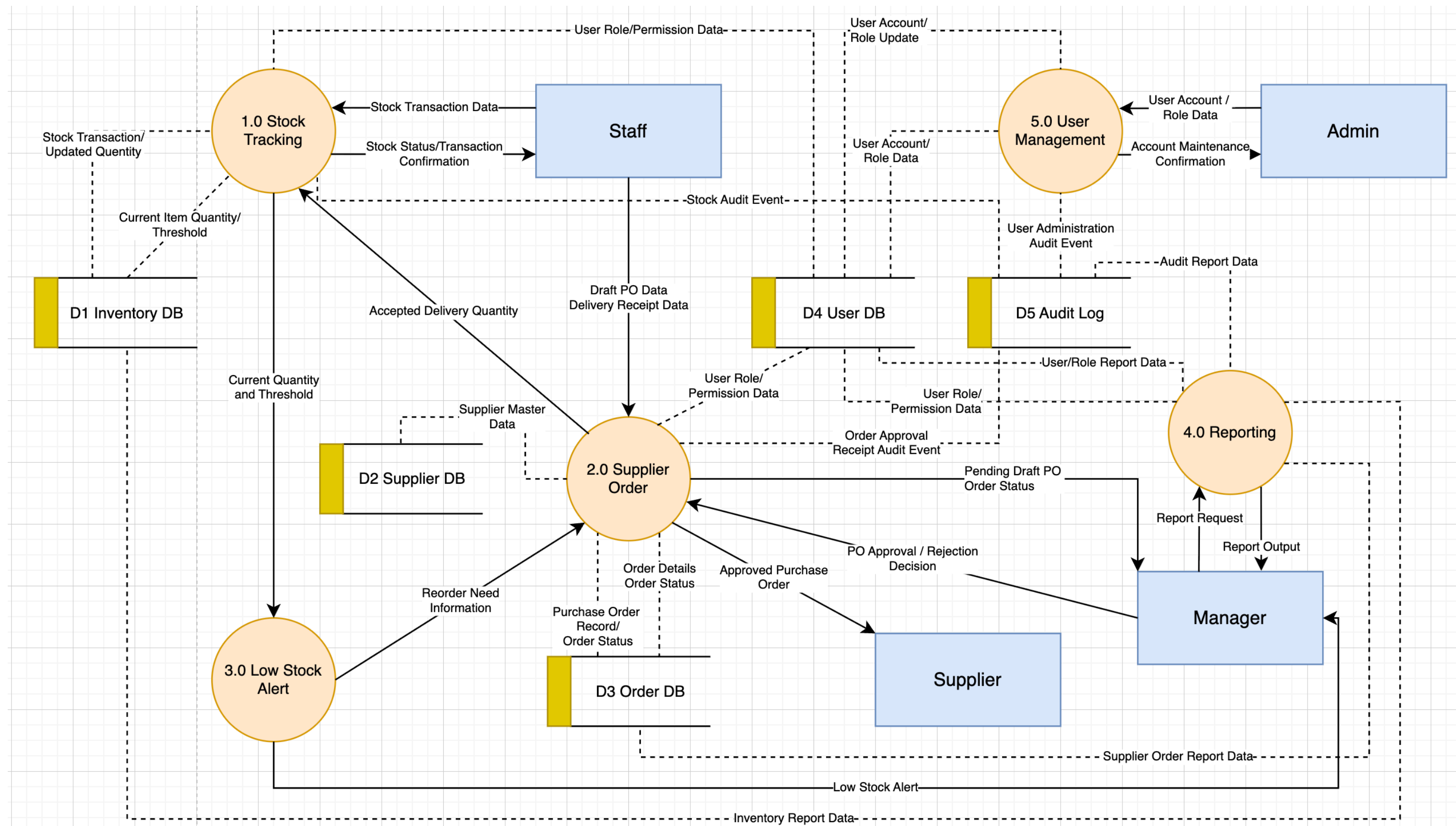
FUNCTIONAL REQUIREMENTS

- The system shall create a low stock alert when the current quantity is less than or equal to the threshold value.
- The system shall allow Manager Users to approve or reject the draft purchase order before releasing to the supplier.

NON-FUNCTIONAL REQUIREMENTS

- The data base supports at least 100000 items, 1000 suppliers and 5 years of transaction history.
- Minimum 99% availability from 8:00 to 18:00 on working days, excluding planned maintenance.

LVL-0 DFD DIAGRAM



AMBIGUITY IN REQUIREMENTS



CONTENT

- Ambiguity occurs when requirements are unclear or can be interpreted in multiple ways.
- This makes development and testing difficult and inconsistent.

EXAMPLES

- “The system shall update stock fast” → “fast” is unclear
- “Managers can view reports when needed” → not specific
- “Supplier receives purchase order” → method not defined

IMPROVED (CLEAR REQUIREMENTS)

- System updates stock within 3 seconds
- Managers can access specific reports after login
- System generates and emails purchase order to supplier

CONFLICT (CONFLICTING REQUIREMENTS)



CONTENT

- Conflicts happen when different stakeholders have opposing needs.
- These cannot be satisfied at the same time without compromise.

EXAMPLES

- Instant alerts vs fewer interruptions
- Open access vs strict security control
- Delete old data vs keep audit history

SOLUTIONS

- Group alerts instead of sending too many
- Allow staff to create orders but require manager approval
- Use archiving instead of deletion

TESTING & QUALITY ASSURANCE

CONTENT

- Testing ensures the system works correctly and meets requirements.

TYPES OF TESTING

- Functional Testing: checks features work correctly
- Negative Testing: ensures invalid actions are rejected
- Edge Testing: checks system behavior at limits

EXAMPLES

- Add stock → quantity increases
- Invalid login → access denied
- Stock = threshold → alert triggered

What is QA Testing and Why is it Important for Software Development?



FUNCTIONAL TEST CASES

ID	Scenario	Input	Expected Output
TC-F-01	Create stock receipt	Staff records Item A receipt of 25 units.	Inventory quantity for Item A increases by 25 and a success message is displayed.
TC-F-02	Issue Stock	Staff issues 5 units of Item B	Inventory quantity for Item B decreases by 5 and the transaction is saved.
TC-F-03	Create draft order	Staff creates a draft purchase order for Supplier S1 with two low-stock items.	A draft order is saved with status Draft and an order number is generated.

DEFECT REPORT

Content:

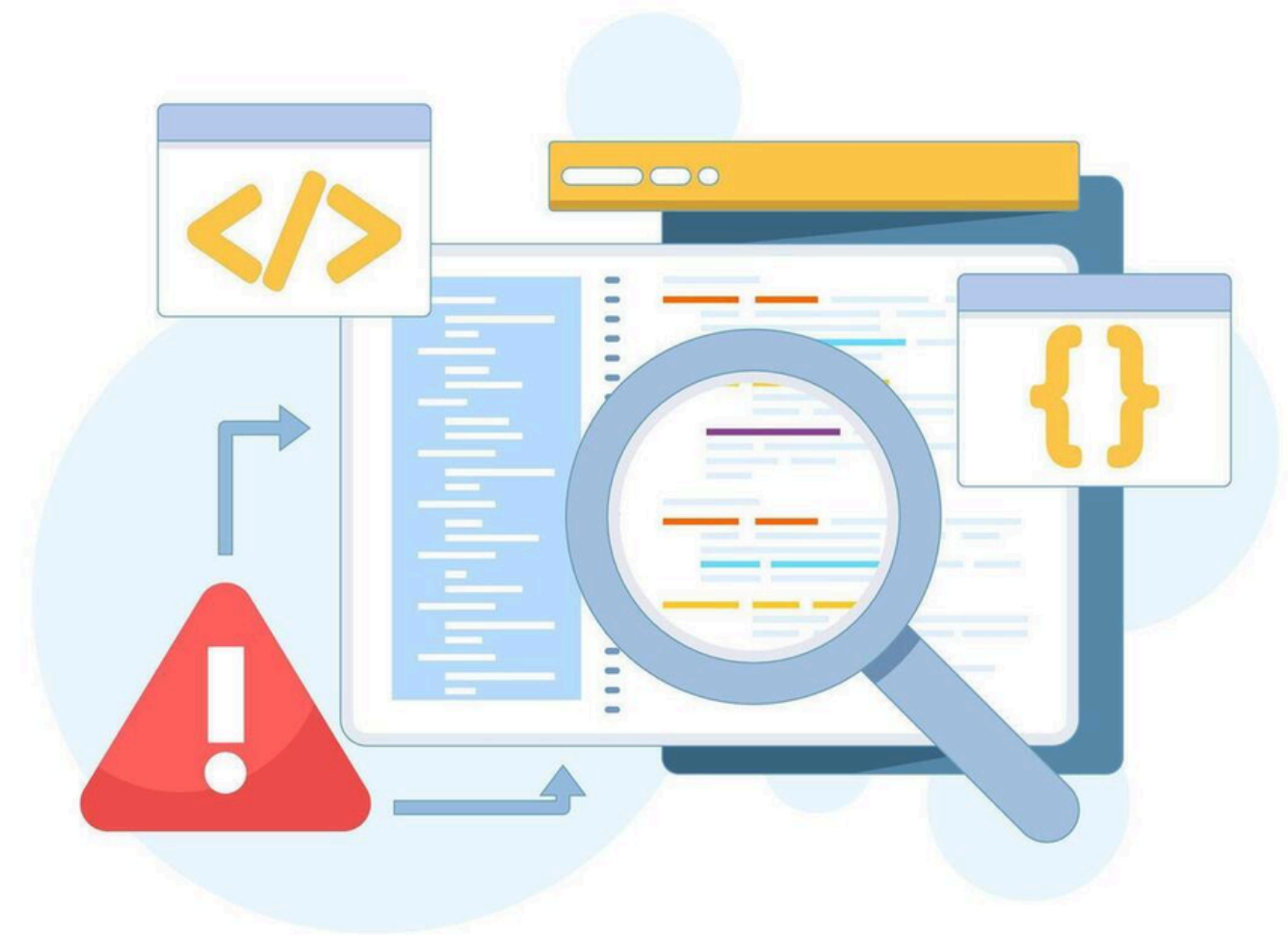
- A defect report records system errors for fixing and tracking.

Defect Example 1:

- Issue: Stock not updated after delivery
- Expected: Stock increases
- Actual: Stock unchanged
- Severity: High

Defect Example 2:

- Issue: No alert when stock = threshold
- Expected: Alert triggered
- Actual: No alert
- Severity: Medium



**THANK
YOU**

